	Accept H_0	Reject H_0
H_0 is true	correct	Type I error
H_0 is false	Type II error	correct

procedure for testing of hypothesis of single Mean.

① Formulate the null hypotheses H_0 such that $\mu = \mu_0$, Then the alternative hypotheses will be any one of the three forms

$H_0 : \mu = \mu_0$

$H_1 : \mu \neq \mu_0$ (two tailed)

OR

$H_1 : \mu < \mu_0$ (left tailed)

OR

$H_1 : \mu > \mu_0$ (Right tailed)

② Specify the level of significance.

Normally $\alpha = 1\%$ OR 5% OR 10%.

③ Select an appropriate test statistic ie,

$$Z = \frac{\bar{x} - \mu}{\text{standard Error}}$$ where

SD of population

$$SE = \frac{\sigma}{\sqrt{n}}, \text{ if } \sigma \text{ is Known}$$

Sample size

OR

$$SE = \frac{s}{\sqrt{n}}, \text{(if } \sigma \text{ is unknown)}$$

$$s^2 = \frac{1}{(n-1)} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

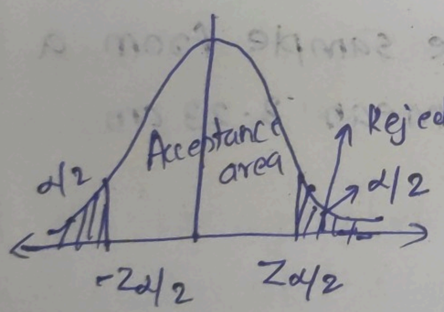
sample mean

4) if $n < 30$, ie, small sample we use student's - t distribution table

if $n \geq 30$ ie, large sample we use standard normal tables

case I

5, For a two tail ie, $(H_1 : \mu \neq \mu_0)$



(i) if $-Z_{\alpha/2} < Z < Z_{\alpha/2}$

$\Rightarrow |Z| < Z_{\alpha/2}$

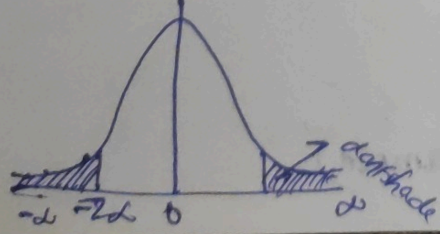
we accept H_0, Reject H_1

(ii) if $Z < -Z_{\alpha/2}$ or $Z > Z_{\alpha/2}$

$\Rightarrow |Z| > Z_{\alpha/2}$

we reject H_0, accept H_1

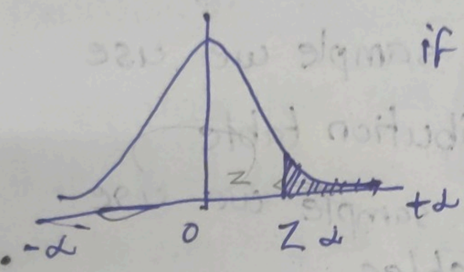
Case II : $H_1 : \mu < \mu_0$ (left tail)



1) if $Z > -Z_\alpha$ Accept H_0

if $Z \leq -Z_\alpha$ Reject H_0

Case III : $H_1 : \mu > \mu_0$ (Right tail).



if $Z < Z_\alpha$ Accept H_0

if $Z > Z_\alpha$ R.

Q A sample of 900 members is found to have a Mean 3.47 cm. Can it be reasonably regarded as a single sample from a large population with mean 3.23 cm and SD 2.31 cm.

Ans $n = 900$

$\bar{x} = 3.47$

$\mu = 3.23$

$\sigma = 2.31$

$H_0 : \mu = 3.23$

$H_1 : \mu \neq 3.23$

Assume $\alpha = 5\%$

$n = 900$ (large sample).

We have normal distribution;

$\alpha = 5\%$, $Z_{\alpha/2} = 1.96$

$$Z = \frac{\bar{x} - \mu}{SE}$$

$$= \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{3.47 - 3.23}{2.31/30}$$

$$= 3.116$$

$|Z| > Z_{\alpha/2}$

So we reject H_0. So the sample cannot be regarded to be taken from the population with mean 3.23 and $\sigma = 2.31$

Q. A stenographer claims that she can take dictation at the rate of 120 ~~wpm~~ words per minute. Can we reject her claim on the basis of 100 trials in which she demonstrates a mean of 116 words and SD of 15 words. Use 5% level of significance.

Ans. $n = 100$

$\mu = 120$

$\bar{x} = 116$ $\quad S = 15$

$H_0 : \mu = 120$

$H_1 : \mu \neq 120$

Given $\alpha = 5\%$ and $n = 100$ (large sample) normal distribution $N(\mu, \sigma)$

$Z_{\alpha/2} = 1.96$

Test statistic:

$$Z = \frac{\bar{x} - \mu}{SE} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{116 - 120}{15/\sqrt{100}}$$

$$= \frac{-4}{15} \times 10 = \frac{-40}{15} \frac{-8}{3}$$

$$= -2.63$$

$$\approx -2.66$$

$|Z| > Z_{\alpha/2}$

Reject H_0.

So her claim is rejected.

Q. A machine is expected to produce nails of length 2cm. A random sample of 25 nails gave an average length 2.1cm and SD 0.25cm. Can it be said that the machine is producing nails as per expected?

Ans $n = 25$

$\mu = 2$

$\bar{x} = 2.1$

$S = 0.25$

$H_0 : \mu = 2$

$H_1 : \mu \neq 2$

Here $n = 25$ (sample space → so we use student's - t distribution)

Degrees of freedom $= n-1$

Here $d.f = n-1 = 25-1 = 24$

At degrees of freedom 24, $Z_{\alpha/2}$

Let $\alpha = 5$, $\alpha/2 = 5/2 = \cancel{0.025}$ $2.5\% = 0.025$

$$Z_{0.025} = 2.064$$

$$Z = \frac{\bar{X}-\mu}{SE} = \frac{\bar{X}-\mu}{S/\sqrt{n}} = \frac{2.1-2}{0.25/\sqrt{25}}$$

$$= 2$$

$|z| < Z_{\alpha/2}$

Accept Ho

10 students are selected at random from a school and their heights are found to be in inches 50, 52, 52, 53, 55, 56, 57, 58, 58, 59. In the light of these data, disuss the suggestion that mean height of students in the school is 54 inches. Use 5% level of significance.

$n = 10, \quad \mu = 54$

$\bar{x} = \frac{\sum x}{n} = \frac{550}{10} = 55$

$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

x	$x - \bar{x}$	$(x-\bar{x})^2$
50	-5	25
52	-3	9
52	-3	9
53	-2	4
55	0	0
56	1	1
57	2	4
58	3	9
58	3	9
59	4	16

$\sum (x - \bar{x})^2 = 86$

$H_0: \mu = 54$, $H_1: \mu \neq 54$

$$s^2 = \frac{1}{10-1} \times 86 = \frac{86}{9} = 9.55$$

$$s = \sqrt{9.55} = 3.09$$

$n = 10$, small sample, use students t distribution

Degrees of freedom = 9

$$Z_{\alpha/2} = Z_{0.025} = 2.262$$

Test statistic,

$$z = \frac{\bar{x} - \mu}{SE} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{55-54}{3.09/\sqrt{10}}$$

$$= 1.023$$

$|z| < z_{\alpha/2}$

Accept H_0.

<u>Hypotheses concerning difference of means of 2 samples</u>

Let $\bar{x}_1$ and $\bar{x}_2$ be the means of 2 samples. Then the test statistic is defined by

$$z = \frac{\bar{x}_1 - \bar{x}_2}{SE} \rightarrow ①$$

(i) if sample is large ① ⇒ ~~Nf~~ standard normal distribution

(ii) if sample is small ① ⇒ student's t distribution with degrees of freedom $n_1 + n_2 - 2$.

(i) $$SE = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

When population variance are known

(ii) $$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

if population variance are unknown.

(iii) If both samples are taken from same population (ie homogenous population).

Then $SD = \sigma \times \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ where

$$\boxed{\sigma^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Q) The means of 2 random samples of sizes 1000 and 2000 are 67.5 and 68 inches respectively. Can the samples be regarded to have been drawn from the same population of SD. 9.5 inches. Test at 5% level of significance.

Ans $n_1 = 1000$, $n_2 = 2000$

$\bar{X}_1 = 67.5$, $\bar{X}_2 = 68$

$\sigma = 9.5$

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

$$SE = \sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$(\hat{p} \pm Z_{\alpha/2}\, SE$

$(\bar{X} \pm Z_{\alpha/2}\, SE)$

$= (\bar{X} - Z_{\alpha/2}\, SE,\ \bar{X} + Z_{\alpha/2}\, SE)$

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{67.5 - 68}{9.5\sqrt{\frac{1}{1000} + \frac{1}{2000}}}$$

$$= \frac{-0.5}{9.5 \times \sqrt{\frac{\cancel{3000}}{\cancel{2000000}}}}$$

$$= \frac{-0.5}{9.5 \times 0.05}$$

$$= \frac{-0.5}{0.36793} = -1.358$$

At $\alpha = 5\%$

$Z_{\alpha/2} = 1.96$

$|Z| < Z_{\alpha/2}$

Accept H_0

$\therefore$ Both the samples are taken from same population.

Q) Given the following data of two distributions

	Mean	SD	sample size
A	100	12	80
B	95	10	70

Test whether the difference b/w the sample mean is significant ?

Ans $H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

$\bar{x}_1 = 100, \quad \bar{x}_2 = 95$

$s_1 = 12 \qquad s_2 = 10$

$n_1 = 80 \qquad n_2 = 70$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{100-95}{\sqrt{\frac{144}{80} + \frac{100}{70}}} = \frac{5}{\sqrt{1.8 + 1.428}}$$

$$= \frac{5}{\sqrt{3.228}} = \underline{\underline{2.78}}$$

Let $\alpha = 5\%$ $\quad Z_{\alpha/2} = 1.96$

$|z| > \cancel{1.96}\ Z_{\alpha/2}$, reject H_0

The difference between sample means is significant.

Q, A ~~sorted~~ certain stimulus administered to each of 12 patients, resulted in the following changes in blood pressure.

Ans 5, 2, 8, −1, 3, 0, −2, 1, 5, 0, 4, 6

Can it be concluded that the stimulus will in general be accompanied by an increase in blood pressure.

Ans $H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 < \mu_2$ (left tail - one tail)

Let difference in blood pressure $\bar{d} - 2.583$

$\Sigma d = 31$, $\bar{x} = \frac{31}{12} = 2.583$

$d = (x_1 - x_2)$	d^2	$d - \bar{x}$	$(d-\bar{x})^2$
5	25		
2	4		
8	64		
−1	1		
3	9		
0	0		
−2	4		
1	1		
5	25		
0	0		
4	16		
6	36		

$$S^2 = \frac{1}{n-1} \Sigma (d_i - \bar{x})^2 = \underline{\underline{9.538}}$$

$$S = \underline{\underline{3.088}}$$

$$\boxed{\bar{x} = \bar{d} = \bar{x}_1 - \bar{x}_2}$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{SE} = \frac{\bar{x}_1 - \bar{x}_2}{\frac{S}{\sqrt{n}}}$$

$$= \frac{2.583}{\frac{3.088}{\sqrt{12}}} = \frac{2.583 \times 3.464}{3.088}$$

$$= \underline{\underline{2.897}}$$

small sample, use student's T distribution.

At $\alpha = 5\% = \frac{5}{100} = 0.05$

$Z_{0.05} =$ at d.f $= n-1 = 11$ is

$|z| > Z_\alpha$, Reject H_0.

Hypotheses concerning single proportion

$H_0 : P = P_0$ → Null hypotheses

$H_1 : P \neq P_0$

OR $H_1 : P > P_0$

OR $H_1 : P < P_0$

} Alternate hypotheses

Test statistic $Z = \dfrac{\hat{p} - P}{SE}$ $\quad SE = \sqrt{\dfrac{PQ}{n}}$

Q A coin is tossed 10,000 times. A head turns up 5195 times. Is the coin unbiased?

$H_0 : P = 1/2$ $\qquad \hat{p} = \dfrac{5195}{10000} = 0.5195$

$H_0 : P \neq 1/2$

$\hat{q} = 1 - p = 0.4805$

$q = 1 - P = 1 - 1/2 = 1/2$

$$Z = \frac{\hat{p} - P}{SE} = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.5195 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{10,000}}}$$

$$= \frac{0.0195}{\sqrt{0.005}}$$

$$= \underline{\underline{3.9}}$$

At 5% level of

$Z_{\alpha/2} = 1.96$

$|z| > Z_{\alpha/2}$

Reject H_0.

Hence the coin is biased.

Q1) In a big city ~~320~~ 325 men out of 600 men were found to be smokers. Does this information support the conclusion that majority of men in the city are smokers? Use 5% level of significance

Ans $H_0 : p = 1/2$

$H_1 : p > 1/2$ (majority)

(Right tailed)

$$\hat{p} = \frac{325}{600}$$

$$p = 1/2 , \quad q = 1/2$$

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.0016 - 0.5}{\sqrt{\frac{0.25}{600}}}$$

$$= \underline{\underline{2.038}}$$

(Right tail +ve
left tail −ve)

$Z_\alpha =$

$$|z| > Z_\alpha$$

$\therefore$ Reject H_0.

$\therefore$ Majority of men are smoker.

Hypotheses concerning difference of proportion

$H_0: P_1 = P_2$

$H_1: P_1 \neq P_2$

OR $H_1: P_1 > P_2$ OR $H_1: P_1 < P_2$

Test statistic:

$$Z = \frac{\hat{P_1} - \hat{P_2}}{SE}, \quad SE =$$

$$SE = \sqrt{\frac{\hat{P_1}\hat{q_1}}{n_1} + \frac{\hat{P_2}\hat{q_2}}{n_2}}$$

$\hat{P_1}, \hat{P_2}$, proportion of 2 samples

If Both the samples are taken from the same (homogenous population) then

$$SE = \sqrt{P_0 q_0 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

where $P_0 = \dfrac{n_1 \hat{P_1} + n_2 \hat{P_2}}{n_1 + n_2}$

Q) In a random sample of 1000 persons, from town A, 400 are found to be consumers of wheat. In a sample, of 800, from town B, 400 are consumers of wheat.

Do these data reveal a significant difference between town A and town B, so far as the proportion of wheat consumers ~~is~~ is concerned ?

Ans $H_0 : P_1 = P_2$, $H_1 : P_1 \neq P_2$ $n_1 = 1000$

$n_2 = 800$

$$\hat{P}_1 = \frac{400}{1000} = \underline{\underline{0.4}} \qquad \hat{P}_2 = \frac{400}{800} = \underline{\underline{0.5}}$$

(Canalyze same category)

Since it is a homogenous population,

$$SE = \sqrt{P_0 q_0 \left({}^1/_{n_1} + {}^1/_{n_2} \right)}$$

where
$$P_0 = \frac{n_1 \hat{P}_1 + n_2 \hat{P}_2}{n_1 + n_2}$$

$$= \frac{1000 \times 0.4 + 800 \times 0.5}{1800}$$

$$= \frac{400 + 400}{1800}$$

$$= \frac{8\cancel{00}}{18\cancel{00}} = \frac{4}{9} = \underline{\underline{\cancel{0.88}\ 0.45}}$$

$q_0 = 1 - P_0 = \cancel{m}\ 0.55$

$SE = \sqrt{0.45 \times 0.55 \left(1/1000 + 1/800 \right)}$

0.001
0.008

$= 0.45 \times 0.55 \times$

$= 0.0235$

Test statistic $z = \dfrac{\hat{P_1} - \hat{P_2}}{SE}$

$= \dfrac{0.4 - 0.5}{0.0235}$

At $\alpha = 5\%$

$= \dfrac{-0.1}{0.0235} = -4.25$

$\cancel{V}$ $Z_{\alpha/2} = 1.96$

$|Z| > Z_{\alpha/2}$

Reject H_0

$\therefore$ Accept alternate hypotheses.

So there is a significant diff. in wheat consumption of 2 towns.

Q, A machine puts out 16 imperfect articles in a sample of 500. After machine is overhauled it puts out three imperfect articles in a batch of 100. Has the machine improved? Test at 5% level of significance

Ans $n_1 = 500$, $n_2 = 100$

$\hat{P_1} = 16/500$ $\hat{P_2} = 3/100$

$= 0.032$

$H_0 : P_1 = P_2$

$H_1 :$ ~~$P_1 < P_2$~~ $P_1 > P_2$ (right tailed)

$$SE = \sqrt{P_0 q_0 \left(1/n_1 + 1/n_2 \right)}$$

$$\cancel{P_0} = \cancel{\frac{P_1 q_1}{n_1}} + \cancel{\frac{P_2 q_2}{n}}$$

$$P_0 = \frac{n_1 \hat{P_1} + n_2 \hat{P_2}}{n_1 + n_2}$$ $SE = 0.019$

$$= 0.0317$$

$q_0 = 1 - P_0 = 1 - 0.0317 =$ ~~0.68~~ 0.9683

$$Z = \frac{\hat{P_1} - \hat{P_2}}{SE} = \frac{\hat{P_1} - \hat{P_2}}{\sqrt{P_0 q_0 (1/n_1 + 1/n_2)}}$$

$= 0.1042$

At $\alpha = 5\%$

$Z_\alpha = 1.65$

$|z| < Z_\alpha$

Accept H_0.

There is no improvement for the machine